
EE1060: Differential Equations and Transform Techniques

Lecture 11: Bifurcations

Topics :

1. Introduction to Bifurcations
 2. Saddle-Node, Transcritical and Pitchfork Bifurcations
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Introduction to Bifurcation Analysis and Examples

$$\frac{di}{dt} = \frac{1}{L} (V_{dc} - Ri)$$

can change w.r.t temperature

$\mu \leftarrow$ varying parameter in the ODE

① if $\mu = L$

$$\frac{di}{dt} = \frac{1}{\mu} (V_{dc} - Ri)$$

$$i^* = \frac{V_{dc}}{R}$$

$$\frac{dy}{dt} = f(y)$$

y^* is a stable equilibrium if $f'(y^*) < 0$.

② if $\mu = R$

$$\frac{di}{dt} = \frac{1}{L} (V_{dc} - \mu i)$$

$$i^* = \frac{V_{dc}}{\mu}$$

$$f'(i) = -\mu/L$$

if $\mu = 0 \rightarrow$ no equilibrium

$\mu > 0 \rightarrow$ exists & is $\frac{V_{dc}}{\mu}$ (stable)

$\mu < 0 \rightarrow$ " " " $\frac{V_{dc}}{\mu}$ (unstable).

goal: to extend it to ODE $\frac{dy}{dt} = f(y, \mu)$

Saddle-node bifurcation

$$\frac{dy}{dt} = \underbrace{\mu - y^2}_{f(y, \mu)} = f(y, \mu)$$

① Equilibrium points $y: f(y) = 0 \Rightarrow y_1^* = \sqrt{\mu} \text{ \& } y_2^* = -\sqrt{\mu}.$

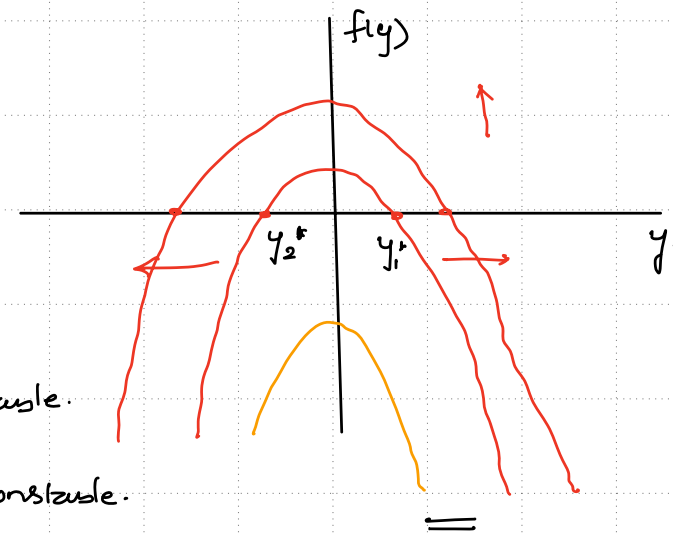
$\mu > 0$: 2 Equilibrium $y_1^* = \sqrt{\mu} \text{ \& } y_2^* = -\sqrt{\mu}$

a) Equilibrium points move apart

b) Stability: $f'(y) = -2y.$

at y_1^* : $f'(y) < 0$, stable.

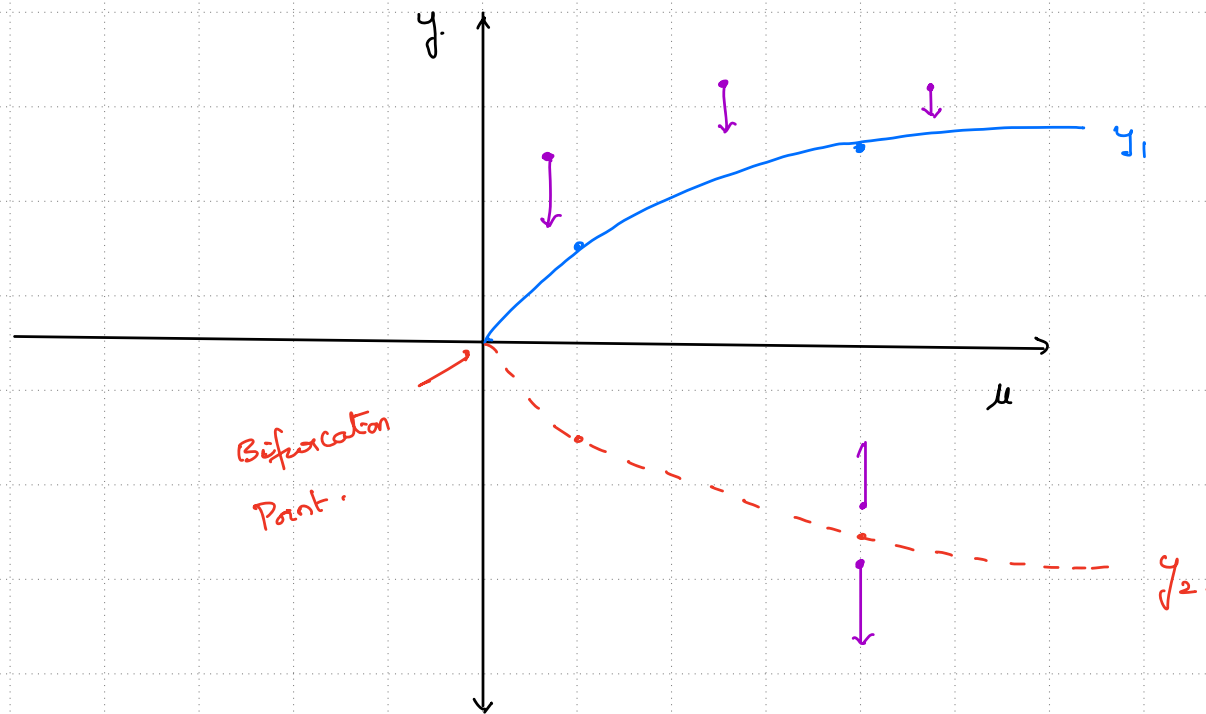
at y_2^* : $f'(y) > 0$, unstable.



$\mu = 0$: 1 Equilibrium point $y_1^* = 0.$

$\mu < 0$: No Equilibrium.

Saddle-node bifurcation - Bifurcation Diagram



Transcritical Bifurcation

$$\frac{dy}{dt} = \mu y - y^2.$$

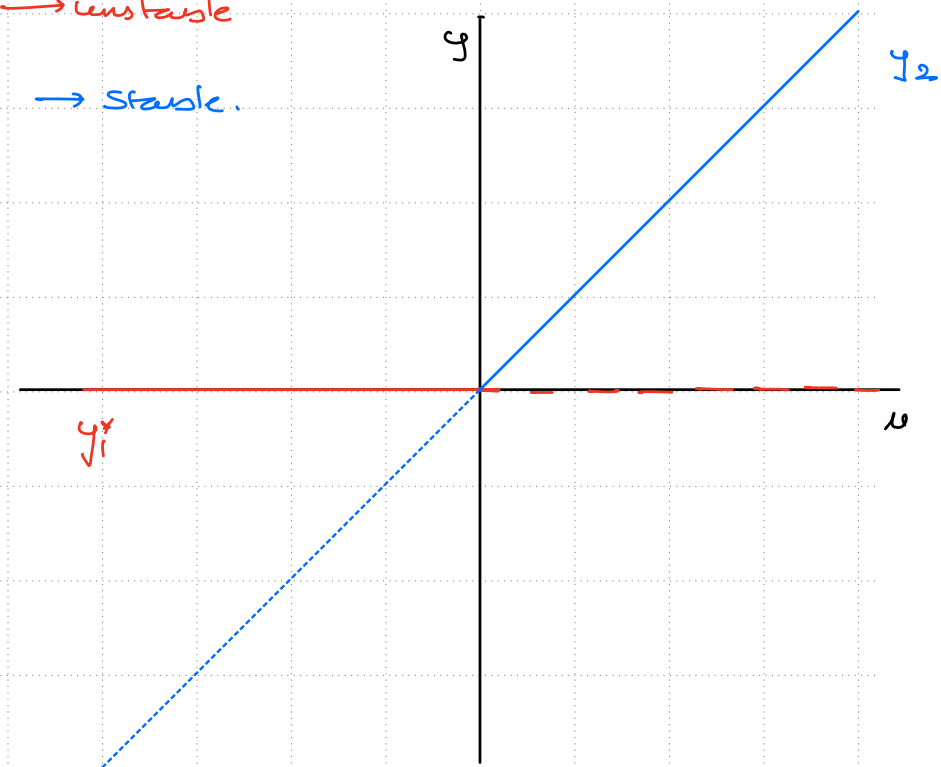
① Equilibrium points : $y_1^* = 0, y_2^* = \mu$

$$f'(y) = \mu - 2y.$$

① $\mu > 0$:
 $y_1^* \leftarrow$ exist $f'(y_1^*) > 0 \rightarrow$ unstable
 $y_2^* \leftarrow$ exist $f'(y_2^*) < 0 \rightarrow$ stable.

② $\mu = 0$: $y_1^* = y_2^* = 0$

③ $\mu < 0$: $y_1^* \rightarrow$ stable.
 $y_2^* \rightarrow$ unstable.



Pitchfork Bifurcation

$$\frac{dy}{dt} = uy - y^3$$

Equilibrium points: $y_1^* = 0$, $y_2^* = \sqrt{u}$, $y_3^* = -\sqrt{u}$

$$f'(y^*) = u - 3(y^*)^2 =$$

$u > 0$: $y_1^* =$ exists unstable.

$y_2^* =$ " stable.

$y_3^* =$ " stable.

$u = 0$: $y_1^* =$
 $y_2^* =$
 $y_3^* =$ } $= 0$.

$u < 0$: $y_1^* =$ stable

$y_2^* =$ don't exist

$y_3^* =$ "

